

Sinhala/Tamil Speaker Verification — Research Progress Report

Benchmark v1: the open-set rebuild, and the architecture / front-end study

SL-SPV project

2026-08-17

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1. Where this sits in the research programme

This report covers **12–17 August 2026**. It is not a standalone study: it is the **v1 open-set benchmark** that the July roadmap named as the arbiter for its own results.

1.1 The arc so far

Phase	Date	What was established
P0–P3	2026-07-03	Benchmark v0 on SLR52 (478 si) + SLR65 (49 ta). Zero-shot table, backend adaptation, PEFT study. Thesis ch. 1–4 and papers/ieee_spl written against these numbers.
Dataset build + QC	2026-08-10	Eight corpora rebuilt into one VoxCeleb-format collection. Full quality audit; two speaker-identity bugs found (SLR127 is 638 speakers not 531; Kathbath 49 not 60). Tamil pool went 50 -> 752 speakers .
Benchmark v1	2026-08-12 – 17	This report. Speaker-disjoint rebuild, staged architecture / loss / front-end / interaction study across six data conditions.

1.2 What v0 concluded, and the caveat it carried

P3 (PEFT fine-tuning) established a clean monotone result — performance improved with encoder freedom:

P3 arm (v0, closed-set)	si EER	ta EER
frozen encoder	3.74	3.13
LoRA (r=8, 1.1 % params)	3.17	2.44
LLRD	2.40	1.81
full fine-tune	1.63	1.30

Three-seed means: full **1.69 / 1.41** vs LoRA **3.73 / 3.11**. Full fine-tuning beat LoRA at every seed, which does *not* reproduce the UniPET-SPK direction.

The P3 log stated its own limit explicitly:

“the 527 eval speakers ARE the training speakers ... This measures representation adaptation, not open-set generalization; **v1 will provide the open-set condition.**” and “Must be reported with the closed-set caveat; **v1 open-set rerun is the arbiter.**”

That rerun is what this report delivers. Everything below is measured on speaker-disjoint splits, so it is the condition v0 could not test.

1.3 What changed between v0 and v1

- **Protocol.** Every corpus rebuilt speaker-disjoint (§3.1). v0’s per-utterance split shared speakers between train and test.
- **Tamil corpus.** v0 used SLR65 (49 speakers). v1 uses **SLR127 (638)**, from the August dataset build — a 13x increase, and the reason Tamil results here are not directly comparable with v0’s.

- **Scope.** v0 studied one architecture family under adaptation. v1 sweeps seven backbones, nine front ends and the front-end x backbone interaction, across six data conditions.
- **Statistics.** v1 adds the paired speaker-clustered bootstrap and a measured power ceiling (§3.2), neither of which v0 had.

1.4 Continuity note on the P3 architecture

P3 used **WavLM with learnable softmax layer weights** over all hidden states (`tools/peft_finetune.py`), which is also what the deployed VoiceID system uses. That design was already correct on the question §4 turns out to be the most consequential one — **which layer the features come from**.

The main trainer’s `SSLFrontendSpeaker` does *not* do this: it reads a single layer, defaulting to the last. Stage A used that default, which is why the 95M-parameter SSL model finished last there (§6) and why the layer investigation began. So v1 did not discover something P3 missed; it **quantified, on open-set data, the thing P3’s architecture had already assumed** — and found the effect is worth 3.3x.

2. Executive summary

54 experiments across six speaker-disjoint data conditions. **50 trained, 24 evaluated, 4 running.**

#	Finding	Evidence
1	Every shipped corpus list is 100 % closed-set. All test speakers also appear in training. Rank-distorting, not merely optimistic — inflation scales with capacity to memorise class centroids.	slr52 478/478, slr127 638/638, slceleb 123/123
2	The SSL front end was reading its worst layer. Speaker information falls <i>monotonically</i> with depth; <code>--ssl_layer -1</code> is the worst available choice.	layer 0 = 22.30 %, layer 12 = 40.33 %; 1.81x / 2.90x worse
3	Tamil’s apparent superiority is largely a channel artifact. SLR127 pools three collection sites; about 48 % of impostor trials are rejected on channel, not identity.	cross-batch 0.475 % vs same-batch 1.298 %
4	Fixing the front end beat every architecture change. Best system is 42 % better than the strongest mel baseline.	2.141 % vs 3.721 % (si)
5	Read speech is about 6x easier than genuine cross-session audio.	read 3.421 % vs wild 21.381 %, same model

Status. All figures are validation EER on single runs. They become claims only after the held-out evaluation and the paired bootstrap complete. A single absolute EER here carries roughly ± 5 –6 pp; only paired contrasts resolve architecture-scale differences.

3. Methodological foundation

3.1 The closed-set problem

A trial score is $s = \cos(f(x_1), f(x_2))$. For a speaker seen in training, f has been explicitly optimised to collapse that speaker onto a learned AAM-softmax centroid w_c , so the between-class margin enforced during training transfers directly into the trial score — the model is evaluated on the objective it was trained on.

The inflation therefore **grows with model capacity**: a 95M SSL front end gains more from the contamination than a 1.4M ResNet. A closed-set ranking of architectures partly ranks *memorisation capacity*, and the two orderings are not the same. This is the concrete mechanism behind the caveat P3 recorded.

`tools/build_splits.py` rebuilds every corpus into speaker-disjoint train / validation / test partitions — 70/10/20 by speaker, stratified by total duration, deterministic from `--seed`, SHA-256 per emitted list. The shipped `lists/` are untouched, so v0 results stay reproducible; they are simply not comparable with v1.

3.2 The power ceiling

Trials sharing a speaker are correlated. With m trials per speaker, S speakers and intra-speaker correlation ρ :

$$n_{\text{eff}} = \frac{mS}{1 + (m-1)\rho} \rightarrow \frac{S}{\rho} \quad \text{as } m \rightarrow \infty$$

Effective sample size is bounded by the **speaker count**, not the trial count. At $S = 91$, $\rho = 0.7$: $n_{\text{eff}} \leq 130$, giving roughly ± 6 pp on a single absolute EER.

Two consequences, both acted on:

1. Trial lists cut **80k -> 20k pairs** — $m \approx 20$ already reaches 98 % of the ceiling, so the remainder was pure evaluation cost.
2. Inference uses a **paired speaker-clustered bootstrap** on EER differences over identical trials, where the speaker-difficulty term cancels. Validated on synthetic ground truth: **2.6x tighter** than absolute EER, correctly detects a known injected difference, correctly finds none between a system and itself. EER matches the repo’s own `tuneThreshold` exactly.

3.3 Controlled factors

Pinned across every run: embedding dim 256, 80 mel bins, 2.0 s train / 3.0 s eval crops, identical MUSAN+RIR augmentation, Adam 1e-3 with 0.95 decay and wd 2e-5, 60 epochs, **per-epoch validation**, cosine scoring with AS-Norm off during training.

Four documented deviations, each recorded in the affected runs’ manifests: VGGVox requires `n_mels=40` architecturally; `ssl_wavlm_ft` uses lr 1e-4 + LLRD (1e-3 destroys a pretrained transformer); `en_full` capped at 10 epochs (measured 341 h for 60); small corpora cap batch at 0.75x speaker count.

Model selection uses **validation trials only**; the test set is scored once, at the validation-selected checkpoint.

3.4 Data conditions

Condition	Corpus	Train speakers	Train utts	Role
si	slr52_sinhala	336	129,042	Sinhala read speech
ta	slr127_tamil	446	63,338	Tamil read speech
si_celeb	slceleb2026	82	31,284	in-the-wild, real sessions
si_pooled	slr52 + slceleb	418	160,326	pooled domains
en_matched	VoxCeleb2 subset	336	47,926	scale-matched English
en_full	VoxCeleb2 -> Vox1-O	5,871	1,065,152	positive control

4. Stage F — the front-end sweep (COMPLETE, 14/14)

The study’s principal result. Backbone, loss, splits and schedule fixed; only the input representation changes.

Front end	si	ta	Family
ssl_mhubert_lw	2.141	0.506	SSL, layer-weighted
ssl_wavlm_lw	2.801	0.587	SSL, layer-weighted
ssl_wavlm_low (layer 0)	3.701	1.113	SSL, single layer
<i>ecapa1024 — mel baseline</i>	<i>3.721</i>	<i>0.810</i>	filterbank
mfcc40d	3.942	1.032	cepstral
mfcc80	4.062	1.093	cepstral
ssl_wavlm_ft (unfrozen)	4.522	1.821	SSL, fine-tuned
ssl_wavlm_mid (layer 6)	4.742	1.579	SSL, single layer
<i>ssl_wavlm — layer 12</i>	<i>7.083</i>	<i>4.837</i>	SSL, last layer

- **Layer choice was worth 3.3x.** Same encoder, same data: 7.083 -> 2.801, purely from how the layers are read.
- **Learned weighting beats every fixed depth** — lw > layer 0 > layer 6 > layer 12. Cost: 13 scalar parameters.
- **Multilingual pretraining pays at equal capacity.** mHuBERT-147 and WavLM are both 94M; mHuBERT wins on both languages, isolating *coverage* from capacity.
- **MFCC is not the disaster predicted** — within about 0.2 pp of log-mel. A pre-registered prediction, falsified.

4.1 The layer probe

Measured directly with **no training**: run the frozen encoder, mean-pool each hidden state, cosine-score 4,000 validation trials.

Layer	si EER %	ta EER %
0 (CNN output)	22.30	11.05
3	29.27	15.78
6	36.50	24.48
9	38.90	31.68
12 (the default)	40.33	32.08

Monotonic in depth, in both languages. This is what masked-prediction pretraining predicts: the objective rewards recovering a masked frame from context, driving upper layers toward phonetic content and treating speaker identity as nuisance.

A pre-registered prediction was falsified. The 12 August log predicted speaker information would peak *mid-stack*, and `ssl_wavlm_mid` was built at layer 6 on that basis. Layer 6 measures 36.5 % — barely better than the last layer. The entry was kept as the mid-curve data point and the falsification recorded, rather than edited away.

4.2 Relation to P3 — and the experiment that would close the loop

P3 found performance **monotone in encoder freedom** (frozen 3.74 -> LoRA 3.17 -> LLRD 2.40 -> full 1.63 on si), closed-set, with a layer-weighted encoder.

v1 finds a **frozen, layer-weighted** encoder (2.141) beating an **unfrozen, single-layer** one (4.522). These are not in direct contradiction, because the two comparisons vary different things:

	layer handling	encoder	si EER
P3 frozen (v0, closed)	layer-weighted	frozen	3.74
P3 full (v0, closed)	layer-weighted	fine-tuned	1.63
v1 ssl_mhubert_lw (open)	layer-weighted	frozen	2.141
v1 ssl_wavlm_ft (open)	single, last	fine-tuned	4.522

The missing cell is layer-weighted + fine-tuned on open-set data — exactly P3’s winning configuration, under v1’s protocol. That single run is the direct open-set arbiter for P3’s headline claim, and it has not been done. It is the highest-value next experiment in the programme (§10).

5. Stage H — does the backbone still matter under SSL features? (2/4)

Stage A varies the backbone with the front end fixed; Stage F varies the front end with the head fixed. Stage H measures the interaction neither can see.

System	si	ta
ssl_mhubert_lw (pooling head)	2.141	0.506
ssl_mhubert_ecapa (full ECAPA backbone)	2.421	0.324

The answer is language-dependent, which is itself the finding. On Tamil the ECAPA backbone helps substantially (0.506 -> 0.324, -36 %); on Sinhala it *hurts* (2.141 -> 2.421, +13 %). At **0.324** %, `ssl_mhubert_ecapa_ta` is the best single result in the programme.

Two `ssl_wavlm_ecapa` runs are still training and will complete the 2x2.

6. Stage A — architecture sweep (COMPLETE)

Architecture	si	ta	Params
ecapa512	3.642	1.032	5.99 M
ecapa1024	3.721	0.810	14.46 M
resnetse34v2	4.242	1.417	7.37 M
resnetse34l	4.362	1.761	1.40 M
mlpmixer	4.582	1.154	7.71 M
vggvox	5.162	1.902	3.64 M
ssl_wavlm (layer 12)	7.083	4.837	94.98 M

ECAPA-512 matches ECAPA-1024 at 40 % of the parameters — the capacity control suggests these corpora are data-limited rather than capacity-limited. **The largest model finished last**, which is what triggered §4.1.

7. Corpus difficulty — the most important caveat

`si_pooled` trains on both Sinhala domains and evaluates them *separately*, so read-speech and cross-session numbers are directly comparable for one model.

Architecture	Read (slr52)	Wild (cross-session)	Ratio
ecapa512	3.421	21.381	6.2x
ecapa1024	3.662	21.521	5.9x
resnetse34v2	4.022	22.563	5.6x
mlpmixer	4.062	23.103	5.7x
resnetse34l	4.522	25.726	5.7x
vggvox	5.202	26.867	5.2x

A about 6x gap between read speech and genuine cross-session audio, for the same model trained on both. This is the single most important caveat on every read-speech number in the programme — including v0’s, which used SLR52 throughout. The architecture *ranking* is nonetheless stable across domains (`ecapa512` leads both), which is reassuring for external validity even where absolute numbers do not transfer.

7.1 The Tamil channel artifact

SLR127 pools three collection sites (ISTL / MILE / MICI). Splitting its impostor trials by whether both sides come from the same site, on `ecapa1024_ta`:

Impostor type	Count	Mean score	EER
cross-batch	4,805	-0.0433	0.475 %
same-batch	5,151	+0.0527	1.298 %
all (as reported)	9,956	—	0.894 %

Cross-batch impostors score *negative* — rejected almost for free because the channel differs, not the speaker. about 48 % of the trial list is that easy kind. Reproduced across architectures at 2.47–3.20x inflation. SLR52 has no such structure, so comparing the two corpora compares protocols as much as languages.

Reporting rule: quote the same-batch figure for SLR127. The like-for-like Sinhala/Tamil gap is then about 3.4x, not about 4.9x.

7.2 English at matched scale

At an identical 336 training speakers, `ecapa1024` scores **7.180 %** on `en_matched` against **3.721 %** on Sinhala. A domain effect, not a language one — VoxCeleb is in-the-wild, SLR52 is clean read speech — reinforcing that corpus condition dominates language in every comparison here.

7.3 A documented negative result

Trained alone on 82 speakers, `si_celeb` reached 23–28 % EER with minDCF of about 0.99 across every architecture. Training worked (loss 7.34 -> 0.66, train accuracy 6 % -> 85 %): the models learned their training speakers and transferred to none. Two design faults — 82 speakers is far too few, and the validation set has **9 speakers** ($n_{\text{eff}} \leq 13$), which cannot measure anything yet selected every checkpoint. Retained as a negative result; its EERs should not be quoted. Its correct use is as an evaluation probe (§7), where it is highly informative.

8. Engineering and infrastructure

8.1 Fourteen bugs found and fixed

Bug	Consequence if unfixed
Checkpoints loaded into the wrong wrapper class	Matched 0 tensors — silently evaluating untrained weights at a believable about 41 % EER
<code>int8</code> labels overflow the error-rate accumulator	Silent corruption of every EER past 127 target trials
GPU memory requirement guessed at 20 GB; measured 0.9–1.5 GB	Every SSL job stranded on one A40 while 15 GB T4s idled
Duplicate scheduling across queues	Two processes writing one checkpoint directory — occurred twice
Reduced-scale runs shared <code>save_path</code> with real runs	Smoke checkpoints silently resumed by full runs
Batch size may not exceed the training-speaker count	Zero batches -> crash on the 82-speaker corpus
Stage F per-entry overrides not forwarded	<code>ssl_wavlm_ft</code> would have trained at lr 1e-3 and wrecked the encoder
AS-Norm cohort resolved against the wrong corpus root	Invalid cohort on every transfer and probe set
English conditions leaked into every evaluation	+57k trials per experiment, about 2.5x cost
ROC plot failed on every run (<code>1 - list</code>)	Fixed via <code>sitecustomize</code> ; trainer untouched

Plus: Stage F front ends leaking into Stage A, manifest clobbering across invocations, `pkill -f` over ssh killing its own shell, and deliberate stops being indistinguishable from node failures.

8.2 Cluster reliability

Four cluster-wide reboots in five days (13th 12:56, 14th 14:54, 16th 07:44, 17th 15:34), plus two nodes dropping off the network independently. Each reboot kills every queue driver on the head node — nothing survives, including tmux.

No training work was lost. Per-epoch checkpointing plus recorder-resume meant every interruption cost at most one epoch. Verified: all five reboot-interrupted runs have gap-free, duplicate-free epoch sequences across the boundary.

8.3 The throughput collapse

Running 7 concurrent jobs collapsed *every* job to 5–16 Hz — including pure-mel runs with no SSL encoder. Diagnosis: 114 dataloader processes across three nodes saturating a single NFS server, with GPU utilisation about 0 % and CPU 86–93 % idle. Capping concurrency at 3 restored **224–1,625 Hz, a 30–100x recovery**. Fewer concurrent jobs finish the work strictly faster here.

9. Current state

Stage	Progress
A — si	7/7 complete
A — ta	7/7 complete
A — si_celeb	7/7 complete
A — si_pooled	6/7
F — front ends	14/14 complete
E — english	7/8
H — hybrids	2/4

50/54 trained, 24/54 evaluated. Four running; the A40 is clearing a 28-run evaluation backlog. One node remains unreachable.

10. Not yet done

1. **Held-out evaluation** — 24 of 54. Everything above is validation EER.
2. **The paired bootstrap** — no contrast has a confidence interval yet. Until then, *no ranking in this report is a claim*.
3. **The P3 arbiter run (§4.2)** — layer-weighted + fine-tuned, open-set. The single experiment that resolves v0’s headline claim under v1’s protocol.
4. **Stages B (loss), C (combined si+ta), D (seeds), M (margin)** — deferred. Stage D matters most: every result here is a single seed.
5. **The mel-scale question** — untested since RawNet3 was removed on request.
6. **Channel-controlled Tamil re-scoring** — the split exists and is applied automatically, but the figures above are still the uncontrolled ones.

11. Recommendations

1. **Report ssl_mhubert_lw as the headline system**, with ssl_mhubert_ecapa for Tamil, quoting the mel baseline alongside so the front-end gain is visible.
2. **Run the §4.2 arbiter next.** It closes the loop with P3 and determines whether the thesis’s PEFT chapter needs revising for the open-set condition.
3. **Use same-batch Tamil figures** in any Sinhala/Tamil comparison.
4. **Quote the cross-session number** as the realistic performance estimate; read-speech EER is about 6x optimistic — this applies retrospectively to v0.
5. **Run Stage D before publishing.** Single-seed differences of a few tenths of a point are not interpretable at this power.
6. **Escalate cluster stability.** Four reboots in five days, two nodes lost.

12. Impact on existing write-ups

- **Thesis ch. 1–4** cite v0 numbers throughout. They remain valid *as stated* — v0 measured representation adaptation, not open-set generalisation — but every quoted EER needs the closed-set caveat made explicit, and §7’s about 6x read-vs-wild factor should temper the real-world framing.
- **papers/ieee_spl** is built on the P3 comparison. Its central claim (full FT beats LoRA) is still awaiting its open-set test (§4.2).
- **VoiceID** deploys p3_full_s42, a layer-weighted fine-tuned WavLM — architecturally the configuration §4 supports. Its published thresholds were calibrated on closed-set trials and should be re-derived on v1 splits.

Full detail: `experiments/README.md`, `experiments/FEATURE_EXTRACTION.md`, and the logs of 2026-08-12 and 2026-08-14. Trainer `trainSpeakerNet.py` unmodified throughout.